NC/CC Separation of Cascades by Inferring the Longitudinal Profile (a, b) from DOM Pulse Time Profiles at Known Energy and Vertex

1 Physics and goal

A cascade-like neutrino event belongs to one of two classes, $c \in \{\mathrm{CC}, \mathrm{NC}\}$:

- ν_e CC: an electromagnetic (EM) shower from the electron, carrying the fraction $1 - y$ of the neutrino energy, plus a hadronic shower carrying y (inelasticity y);
- NC (all flavours): a hadronic shower only; the outgoing neutrino is invisible.

Hadronic showers develop more slowly and fluctuate more than EM showers of the same energy. The mean profiles used by PPC (CWLR parameterization, E in GeV, $X_0 = 0.392\,\mathrm{m}$) are

$$g(s; a, b) = \frac{s^{a-1} e^{-s/b}}{\Gamma(a)\, b^a}, \qquad \begin{aligned} a_{\mathrm{EM}} &= 2.018 + 0.632 \ln E, & b_{\mathrm{EM}} &= X_0/0.632 = 0.619\ \mathrm{m}, \\ a_{\mathrm{h}} &= 1.584 + 0.419 \ln E, & b_{\mathrm{h}} &= X_0/0.338 = 1.157\ \mathrm{m}. \end{aligned} \tag{1}$$

E [GeV]	$m_{\mathrm{EM}} = ab$ [m]	$\sigma_{\mathrm{EM}} = \sqrt{a}\, b$ [m]	m_{h} [m]	σ_{h} [m]
10^3	3.95	1.56	5.18	2.45
10^4	4.85	1.73	6.30	2.70
10^5	5.76	1.89	7.41	2.93

At fixed energy, NC showers are about 30% longer and 50% wider on average. ν_e CC showers lie between the two curves, closer to EM, because typically $y \lesssim 0.3$. Event-to-event fluctuations of (a, b) overlap the two classes. That overlap, together with how precisely (a, b) can be recovered from the light, limits the separation.

Strategy. Solve the inverse problem $\mathcal{D} \mapsto p(a, b \mid \mathcal{D}, \boldsymbol{\kappa})$ (Section 3), then classify from the inferred profile (Section 4).

2 Inference problem

Definition 1 (Known quantities, unknowns, data)**.**

- Known: $\boldsymbol{\kappa} = (E,\ \boldsymbol{v},\ t_v)$, i.e. the visible energy, vertex and vertex time.
- Unknown target: $\boldsymbol{y} = (\ln a,\ \ln b) \in \mathbb{R}^2$.
- Nuisance: the shower axis $\hat{\boldsymbol{u}} \in S^2$. It is dropped if the direction is also known.

- Data: $\mathcal{D} = \{(\boldsymbol{r}_d,\ \boldsymbol{n}_d)\}_{d=1}^{N}$, where $\boldsymbol{n}_d = (n_{d1}, \ldots, n_{dK})$ is the photoelectron time profile of DOM d in K bins.

Remark 1 (Which energy is known)**.** *E must be the* EM-equivalent visible *energy, i.e. what a reconstruction returns. If E were the true deposited energy, the lower light yield of hadronic showers (about 20–30% less light) would separate NC from CC through the total charge alone, independently of the profile shape.*

Forward model. With $\boldsymbol{r}(s) = \boldsymbol{v} + s\hat{\boldsymbol{u}}$ and $t_e(s) = t_v + s/c$, the expected count in bin k of DOM d is

$$\mu_{dk}(\boldsymbol{y}, \hat{\boldsymbol{u}}; \boldsymbol{\kappa}) = E \int_0^\infty ds\ g(s; a, b) \int_{\tau_k}^{\tau_{k+1}} dt\ G_d\big(\boldsymbol{r}(s), \hat{\boldsymbol{u}};\ t - t_e(s)\big)\ +\ \nu_{dk}, \tag{2}$$

where G_d is the photon Green's function (PE per GeV per unit delay) and ν_{dk} the noise.

Assumption 1 (Conditional independence)**.** Given $(\boldsymbol{y}, \hat{\boldsymbol{u}}, \boldsymbol{\kappa})$, the counts n_{dk} are independent Poisson variables. Hence

$$\ell(\boldsymbol{y}, \hat{\boldsymbol{u}}) := \ln p(\mathcal{D} \mid \boldsymbol{y}, \hat{\boldsymbol{u}}, \boldsymbol{\kappa}) = \sum_{d=1}^{N} \sum_{k=1}^{K} \big[n_{dk} \ln \mu_{dk} - \mu_{dk} - \ln n_{dk}!\big]. \tag{3}$$

Target of inference. The posterior of the profile given the pulses, with the direction marginalized:

$$p(\boldsymbol{y} \mid \mathcal{D}, \boldsymbol{\kappa}) = \frac{\int e^{\ell(\boldsymbol{y}, \hat{\boldsymbol{u}})}\, p(\boldsymbol{y}, \hat{\boldsymbol{u}} \mid \boldsymbol{\kappa})\, d\hat{\boldsymbol{u}}}{\iint e^{\ell(\boldsymbol{y}', \hat{\boldsymbol{u}})}\, p(\boldsymbol{y}', \hat{\boldsymbol{u}} \mid \boldsymbol{\kappa})\, d\hat{\boldsymbol{u}}\, d\boldsymbol{y}'}. \tag{4}$$

The prior $p(\boldsymbol{y} \mid \boldsymbol{\kappa}) = \sum_c \pi_c\, p(\boldsymbol{y} \mid c, \boldsymbol{\kappa})$ is the class mixture of the simulated sample, with class fractions π_c.

Labels. For each simulated event e (CC or NC), $\boldsymbol{y}_e$ is the Gamma fit to the charged-track-length profile of *all* secondaries, the EM and hadronic parts together:

$$(a_e, b_e) = \arg\max_{a,b}\ \sum_i w_i \Big[(a-1)\ln s_i - \frac{s_i}{b} - \ln\Gamma(a) - a\ln b\Big], \qquad s_i = (\boldsymbol{r}_i - \boldsymbol{v}) \cdot \hat{\boldsymbol{u}}_e. \tag{5}$$

3 Stage 1: inferring (a, b)

3.1 Solution A: Neptune, conditional NLL minimization

Inputs relative to the known vertex. Because $\boldsymbol{\kappa}$ is known, express every hit relative to it:

$$\boldsymbol{\xi}_j = \big(\boldsymbol{r}_j - \boldsymbol{v},\ t_j - t_v\big), \qquad t_j^{\mathrm{res}} = t_j - t_v - \frac{n_g}{c}\|\boldsymbol{r}_j - \boldsymbol{v}\|, \tag{6}$$

where t_j^{res} is added to the per-DOM features $\boldsymbol{f}_j$. With vertex-relative inputs the model does not have to learn translation invariance. With the 4-D rotary encoding, the attention scores depend only on differences $\boldsymbol{\xi}_l - \boldsymbol{\xi}_k$ in any case.

Model. Let $\boldsymbol{h}_\phi(\mathcal{D}) = \mathrm{Pool} \circ \mathrm{Enc} \circ \mathrm{Tok}(\{(\boldsymbol{\xi}_j, \boldsymbol{f}_j)\})$ be the Neptune embedding. The energy is appended before the head, $\boldsymbol{c}_\phi = \boldsymbol{h}_\phi(\mathcal{D}) \oplus [\ln E,\ \ln(1 + Q_{\mathrm{tot}})]$, and the head outputs a bivariate Gaussian:

$$q_\phi(\boldsymbol{y} \mid \mathcal{D}, \boldsymbol{\kappa}) = \mathcal{N}\big(\boldsymbol{y};\ \boldsymbol{\mu}_\phi(\boldsymbol{c}_\phi),\ L_\phi L_\phi^\top\big), \qquad (L_\phi)_{ii} = \mathrm{softplus}(\cdot) + \varepsilon. \tag{7}$$

Objective. Minimize the expected KL divergence from the true posterior (**??**). Up to a constant, this is the expected conditional NLL, estimated on M simulated events drawn from both classes in proportions π_c:

$$\hat{\phi} = \arg\min_{\phi} \frac{1}{M} \sum_{e=1}^{M} \Big[\tfrac{1}{2} \big\| L_{\phi,e}^{-1} (\boldsymbol{y}_e - \boldsymbol{\mu}_{\phi,e}) \big\|^2 + \ln(L_{\phi,e})_{11} + \ln(L_{\phi,e})_{22} \Big] + \frac{\lambda}{2} \|\phi\|^2. \tag{8}$$

Proposition 1. *For every $(\mathcal{D}, \boldsymbol{\kappa})$, the population minimizer of (**??**) (with $\lambda \to 0$ and unlimited capacity) is $\boldsymbol{\mu}^* = \mathbb{E}[\boldsymbol{y} \mid \mathcal{D}, \boldsymbol{\kappa}]$ and $\Sigma^* = \mathrm{Cov}[\boldsymbol{y} \mid \mathcal{D}, \boldsymbol{\kappa}]$, whether or not the posterior is Gaussian.*

Proof. $\mathbb{E}_{\boldsymbol{y}|\mathcal{D}}[-\ln \mathcal{N}(\boldsymbol{y}; \boldsymbol{\mu}, \Sigma)] = \frac{1}{2} \operatorname{tr}(\Sigma^{-1} C) + \frac{1}{2} (\boldsymbol{m} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{m} - \boldsymbol{\mu}) + \frac{1}{2} \ln \det \Sigma + \text{const}$, with $\boldsymbol{m}$ and C the posterior mean and covariance. The second term is minimized at $\boldsymbol{\mu} = \boldsymbol{m}$. The rest is strictly convex in $P = \Sigma^{-1}$, with its stationary point at $P = C^{-1}$. □

Output. One forward pass gives $\hat{\boldsymbol{y}} = \boldsymbol{\mu}_{\hat{\phi}}(\mathcal{D}, \boldsymbol{\kappa})$ and $\hat{\Sigma} = \Sigma_{\hat{\phi}}(\mathcal{D}, \boldsymbol{\kappa})$. A conditional normalizing flow can replace the Gaussian head when the posterior is visibly non-Gaussian.

3.2 Solution B: direct likelihood maximization

With a photon kernel retrained on hadronic and EM showers of varying (a, b), so that it approximates the a, b-dependent Green's function in (**??**), and with $\boldsymbol{\kappa}$ fixed, only 2 parameters (4 if the direction is free) remain:

$$(\hat{\boldsymbol{y}}, \hat{\boldsymbol{u}}) = \arg\max_{\boldsymbol{y}, \hat{\boldsymbol{u}}} \ \ell(\boldsymbol{y}, \hat{\boldsymbol{u}}), \qquad \hat{\Sigma} = \big[-\nabla_{\boldsymbol{y}}^2 \, \ell_p(\boldsymbol{y}) \big]_{\hat{\boldsymbol{y}}}^{-1}, \qquad \ell_p(\boldsymbol{y}) = \max_{\hat{\boldsymbol{u}}} \ell(\boldsymbol{y}, \hat{\boldsymbol{u}}), \tag{9}$$

where ℓ_p is the profile log-likelihood, with the direction maximized out.

4 Stage 2: NC/CC decision from the inferred profile

Optimal test. By the Neyman–Pearson lemma, at fixed $\boldsymbol{\kappa}$ the most powerful test of CC against NC thresholds the likelihood ratio of the full data, $\Lambda^*(\mathcal{D}) = \ln p(\mathcal{D} \mid \mathrm{CC}, \boldsymbol{\kappa}) - \ln p(\mathcal{D} \mid \mathrm{NC}, \boldsymbol{\kappa})$. The profile-based approach replaces $\mathcal{D}$ by the profile. It is exact under the following assumption.

Assumption 2 (Profile sufficiency)**.** $p(\mathcal{D} \mid \boldsymbol{y}, \hat{\boldsymbol{u}}, c, \boldsymbol{\kappa}) = p(\mathcal{D} \mid \boldsymbol{y}, \hat{\boldsymbol{u}}, \boldsymbol{\kappa})$: once the profile is fixed, the pulses carry no further information about the class.

Proposition 2 (Class posterior from the profile posterior)**.** *Under profile sufficiency,*

$$P(c \mid \mathcal{D}, \boldsymbol{\kappa}) = \pi_c \int q(\boldsymbol{y} \mid \mathcal{D}, \boldsymbol{\kappa}) \, \frac{p(\boldsymbol{y} \mid c, \boldsymbol{\kappa})}{\sum_{c'} \pi_{c'} \, p(\boldsymbol{y} \mid c', \boldsymbol{\kappa})} \, d\boldsymbol{y}, \tag{10}$$

where $q = p(\boldsymbol{y} \mid \mathcal{D}, \boldsymbol{\kappa})$ is the stage-1 posterior, computed with the same mixture prior.

Proof. $P(c \mid \mathcal{D}) = \int P(c \mid \boldsymbol{y}, \mathcal{D}) \, p(\boldsymbol{y} \mid \mathcal{D}) \, d\boldsymbol{y}$. By sufficiency $P(c \mid \boldsymbol{y}, \mathcal{D}) = P(c \mid \boldsymbol{y})$, and by Bayes' rule $P(c \mid \boldsymbol{y}) = \pi_c \, p(\boldsymbol{y} \mid c) / p(\boldsymbol{y})$. □

Computation. The class-conditional densities $p(\boldsymbol{y} \mid c, \boldsymbol{\kappa})$ are estimated from the labelled simulation in energy bins, for example as Gaussians $\mathcal{N}(\boldsymbol{m}_c(E), S_c(E))$. Equation (**??**) is then evaluated by Monte Carlo with $\boldsymbol{y}^{(r)} \sim q$:

$$\hat{P}(c \mid \mathcal{D}, \boldsymbol{\kappa}) = \frac{\pi_c}{R} \sum_{r=1}^{R} \frac{p(\boldsymbol{y}^{(r)} \mid c, \boldsymbol{\kappa})}{\sum_{c'} \pi_{c'} \, p(\boldsymbol{y}^{(r)} \mid c', \boldsymbol{\kappa})}. \tag{11}$$

Gaussian closed form. When the stage-1 prior is broad compared with the likelihood, q is approximately the normalized likelihood, and $\hat{\boldsymbol{y}}$ behaves as a measurement $\hat{\boldsymbol{y}} \mid \boldsymbol{y} \sim \mathcal{N}(\boldsymbol{y}, \hat{\Sigma})$. Convolving with the class densities gives

$$\Lambda(\mathcal{D}) = \ln \mathcal{N}\big(\hat{\boldsymbol{y}};\ \boldsymbol{m}_{\mathrm{CC}},\ S_{\mathrm{CC}} + \hat{\Sigma}\big) - \ln \mathcal{N}\big(\hat{\boldsymbol{y}};\ \boldsymbol{m}_{\mathrm{NC}},\ S_{\mathrm{NC}} + \hat{\Sigma}\big), \qquad \text{decide CC if } \Lambda(\mathcal{D}) > \eta. \quad (12)$$

This statistic automatically down-weights events whose profile is poorly measured (large $\hat{\Sigma}$). The threshold η sets the working point on the ROC curve.

Marginal-likelihood version (Solution B). Without passing through q,

$$\ln p(\mathcal{D} \mid c, \boldsymbol{\kappa}) = \ln \int e^{\ell_p(\boldsymbol{y})}\, p(\boldsymbol{y} \mid c, \boldsymbol{\kappa})\, d\boldsymbol{y} \approx \ell_p(\hat{\boldsymbol{y}}_c) + \ln p(\hat{\boldsymbol{y}}_c \mid c, \boldsymbol{\kappa}) + \ln 2\pi - \tfrac{1}{2} \ln \det H_c, \quad (13)$$

where $\hat{\boldsymbol{y}}_c$ is the maximizer of the integrand and H_c the Hessian of its negative log at $\hat{\boldsymbol{y}}_c$ (Laplace approximation). The test statistic is $\Lambda = \ln p(\mathcal{D} \mid \mathrm{CC}, \boldsymbol{\kappa}) - \ln p(\mathcal{D} \mid \mathrm{NC}, \boldsymbol{\kappa})$.

5 Limits and evaluation

Information loss. Because $\hat{\boldsymbol{y}}$ is a function of $\mathcal{D}$, the data-processing inequality gives

$$I(c;\ \hat{\boldsymbol{y}} \mid \boldsymbol{\kappa}) \ \le\ I(c;\ \mathcal{D} \mid \boldsymbol{\kappa}). \quad (14)$$

Compressing the pulses to an estimated profile can only lose class information. Under profile sufficiency, $c \to \boldsymbol{y} \to \mathcal{D}$ is a Markov chain, so in addition $I(c; \mathcal{D} \mid \boldsymbol{\kappa}) \le I(c; \boldsymbol{y} \mid \boldsymbol{\kappa})$: the true profile bounds what the pulses can reveal. Without sufficiency, $\mathcal{D}$ can carry class information that (a, b) does not. The profile route is therefore interpretable but not guaranteed optimal. Hadronic showers differ from EM ones in more than (a, b): larger light-yield fluctuations, delayed light from neutron captures, and muons from meson decays.

Benchmark. Train the same Neptune encoder as a direct classifier, $P_\phi(\mathrm{CC} \mid \mathcal{D}, \boldsymbol{\kappa})$, with binary cross-entropy (the existing `neutrino_classification` task). Its ROC curve estimates the achievable upper bound. The gap between it and the profile-based ROC measures how much class information lies outside (a, b).

Metrics, in bins of E.

- Separation: AUC, and the NC rejection at fixed CC efficiency.
- Stage-1 calibration: the squared Mahalanobis distance $(\boldsymbol{y}_e - \hat{\boldsymbol{y}}_e)^\top \hat{\Sigma}_e^{-1} (\boldsymbol{y}_e - \hat{\boldsymbol{y}}_e)$ should follow χ^2_2.
- Intrinsic overlap: the separation using the *true* $\boldsymbol{y}_e$, i.e. with $\hat{\Sigma} = 0$ and the true labels. This is the best any profile-based method can do.